TSMC018 PDK Utility Usage Guide

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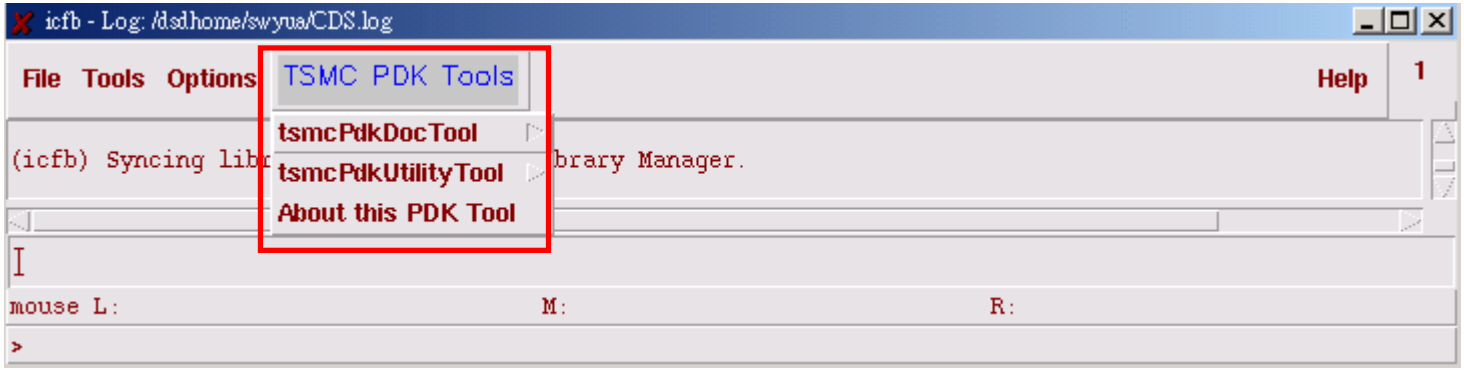
- **tsmcPdkDocTool**
  - Document Center Viewer
- **tsmcPdkToolAbout**
- **tsmcPdkUtility Tool**
  - VXL Fold and Chain setting
  - Switch instances between basic <-> mac
  - Update design(instance) CDF
  - Add MOM license key
  - Create GuardRing utility
  - Layout vs. Layout
  - ZoomIn(sch/lay) with history
  - File Browser & Viewer



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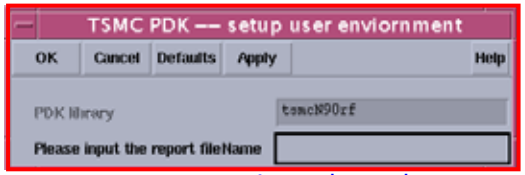
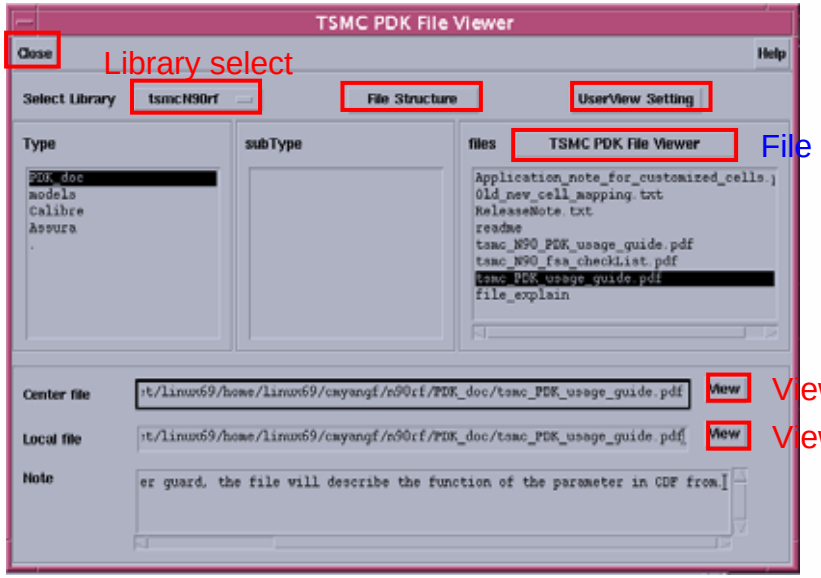
Document Center Viewer :

- Sometimes TSMC PDK user will use symbolic link to use the PDK, user will encounter a problem - documents and technology file path missing. TSMC provide Document Center Viewer function to solve this problem. The function will re-connect the document directory and replace the default path setup.



Input the path to save setup report



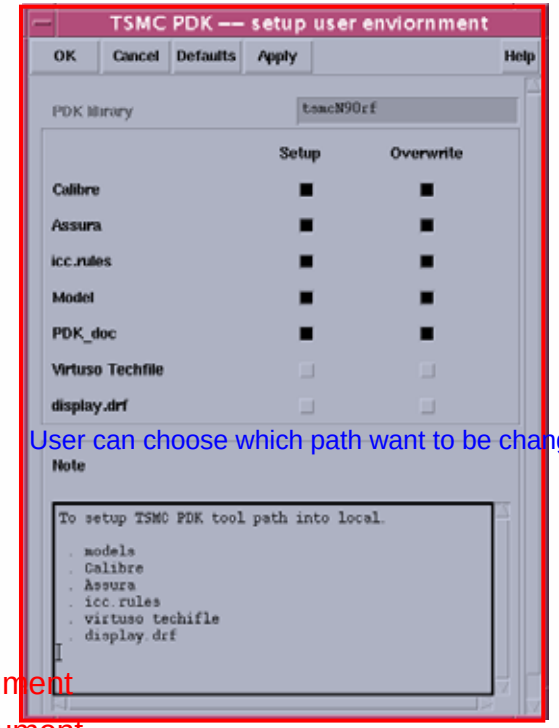
Close window

Library select

File Browser

View PDK document

View Local document



User can choose which path want to be change

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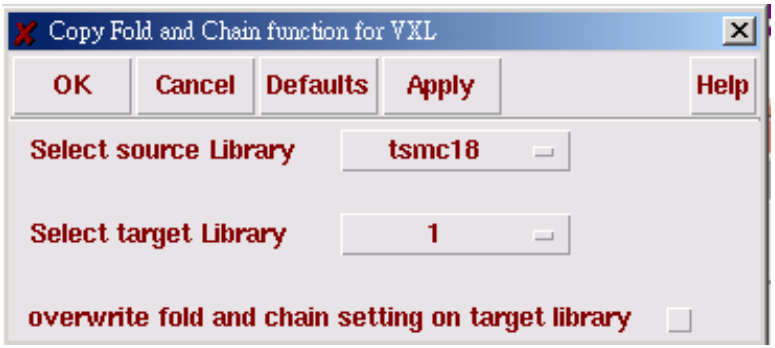
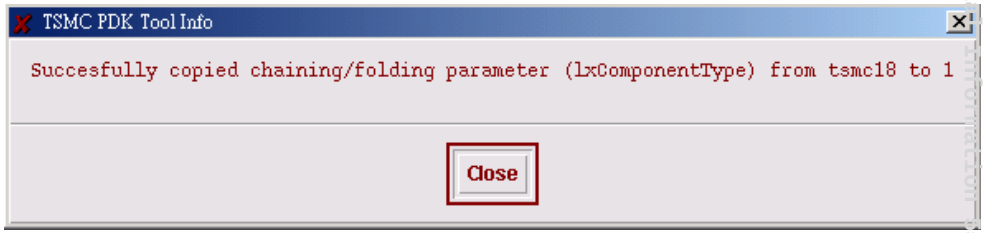
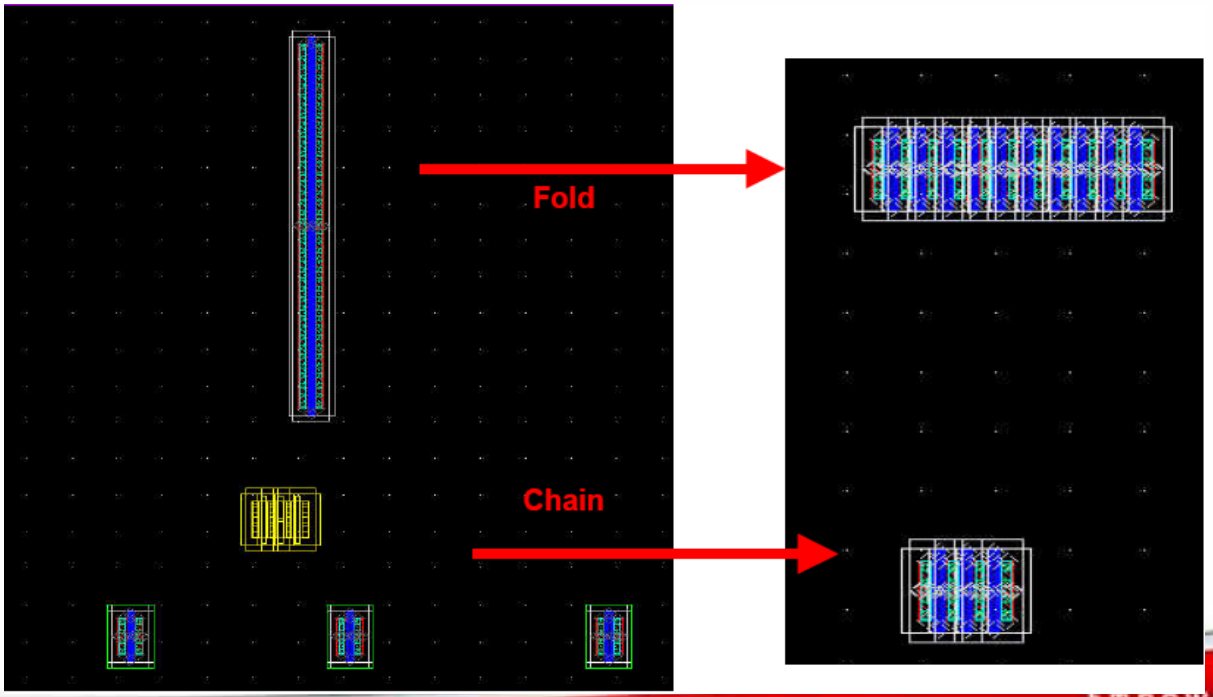
- **VXL Fold and Chain setting :**
  - The function of VXL Fold and Chain setting is an utility function to fold and chain.







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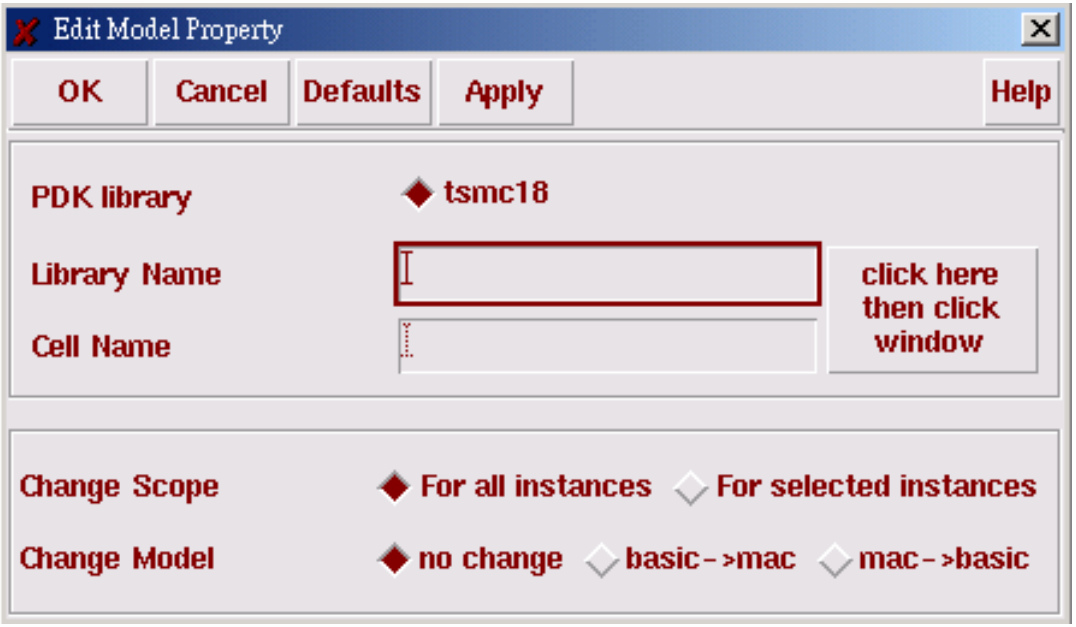
- **Switch instances between basic <-> mac :**
  - The function of Switch instances between basic <-> mac will switch instance between basic and MACRO one.



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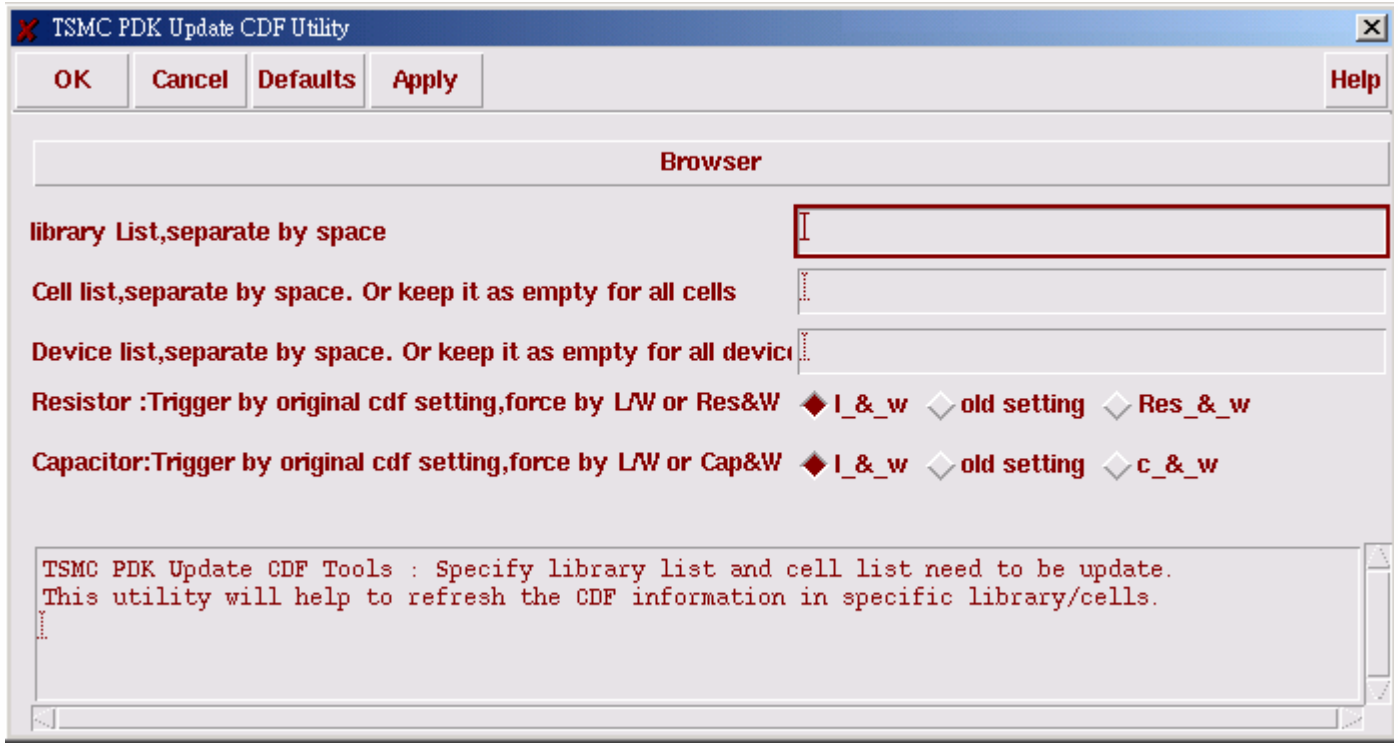
- **Update design(instance) CDF:**
  - The function of Update design(instance) CDF will trigger TSMC CDF parameter update.



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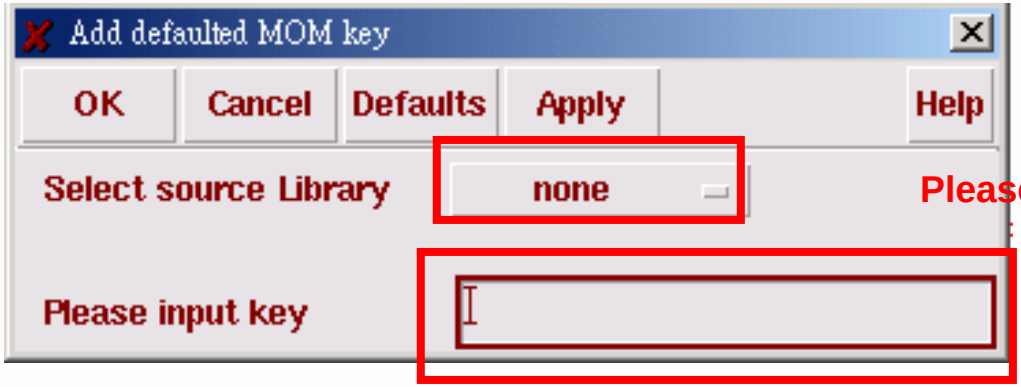
Add MOM license key :

- The function of Add MOM license key will need license key. Please contact TSMC`s AE.



Please Select TSMC PDK Library

Please enter the license key

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Back to Overview

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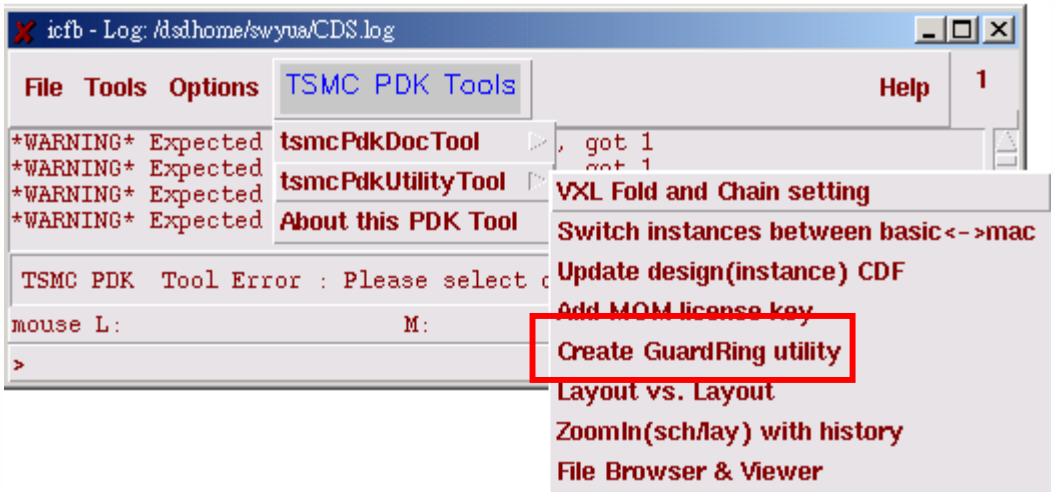
- **Create GuardRing utility :**
  - Create GuardRing utility – TSMC Guard-Ring utility allow users to easily create guard-ring with flexibility. For more information, please refer to “Guard Ring Utility Usage Guide”.



Guard-Ring utility for layout view

. DRC clean

. Create ring path by mouse draw

. Create ring box by mouse click

. Create ring box outside of the each selected instances

. Create ring box outside of the whole selection

. Flexibility for the ring box by specified distance

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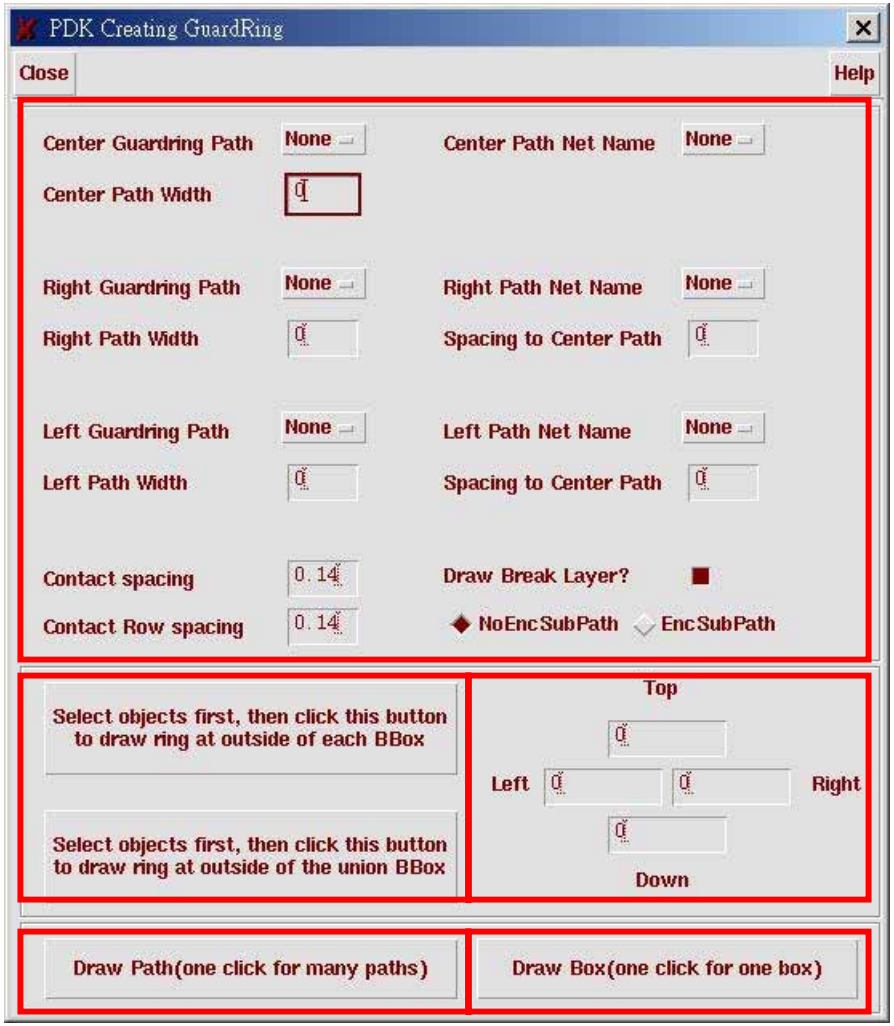
- **Create GuardRing utility :**



Guard-Ring setting

. Psub/NWELL
. Path width
. Contact spacing
. Multi-Path

Guard-Ring box setting

. Separated box
. Union box

Guard-Ring drawing by hand

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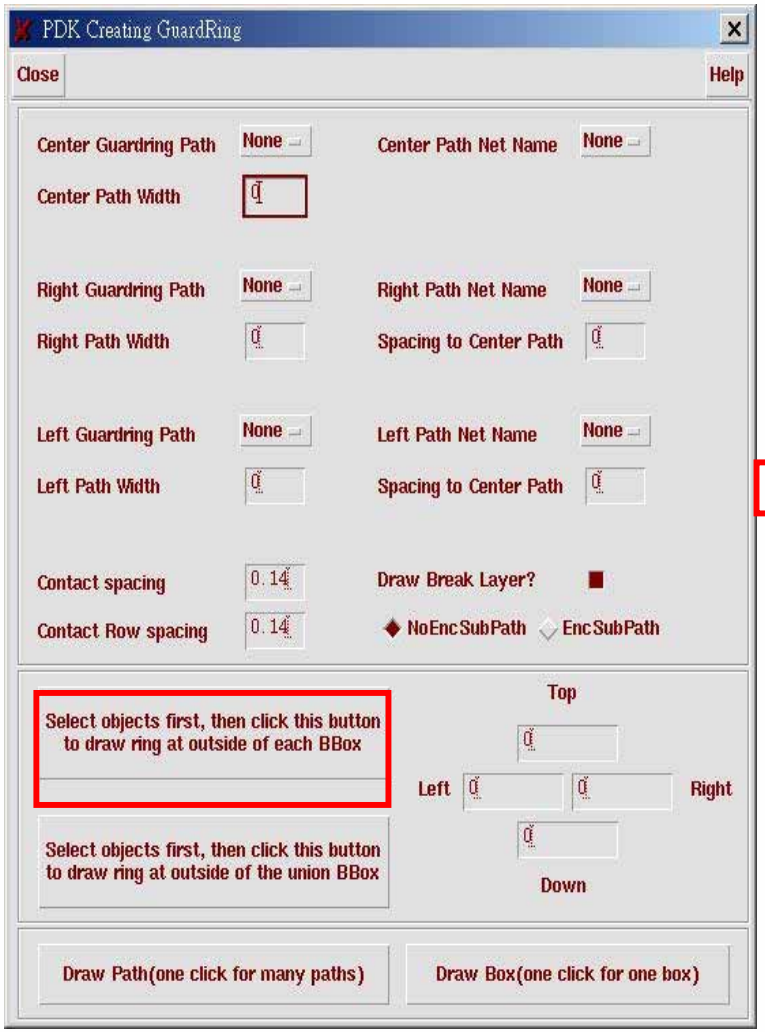
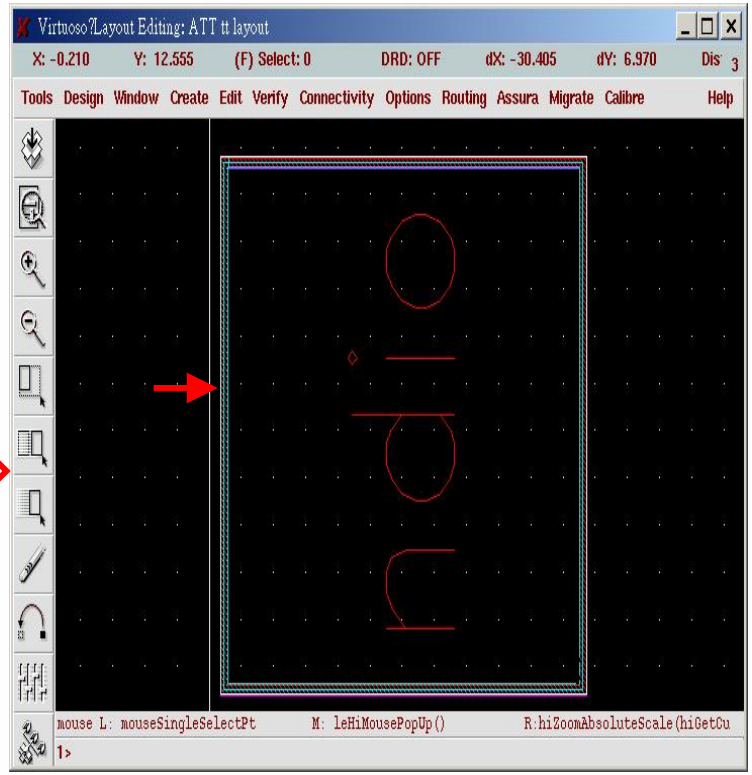
- **Create GuardRing utility :**





Create Guard-Ring outside of the instance

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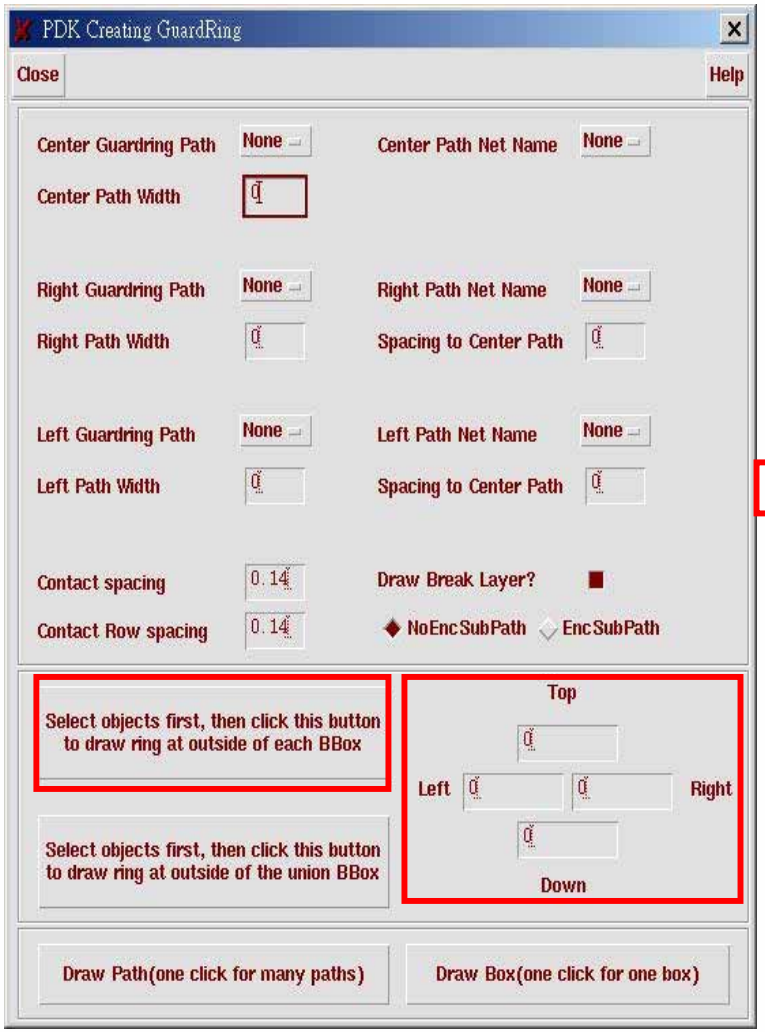
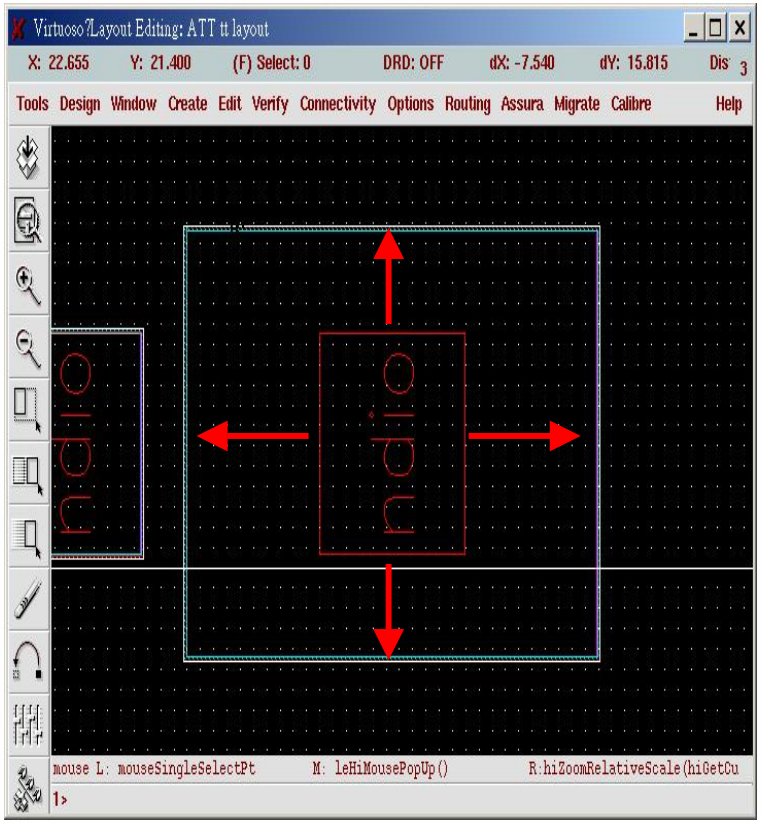
- **Create GuardRing utility :**





Create Guard-Ring outside of the instance by specified distance

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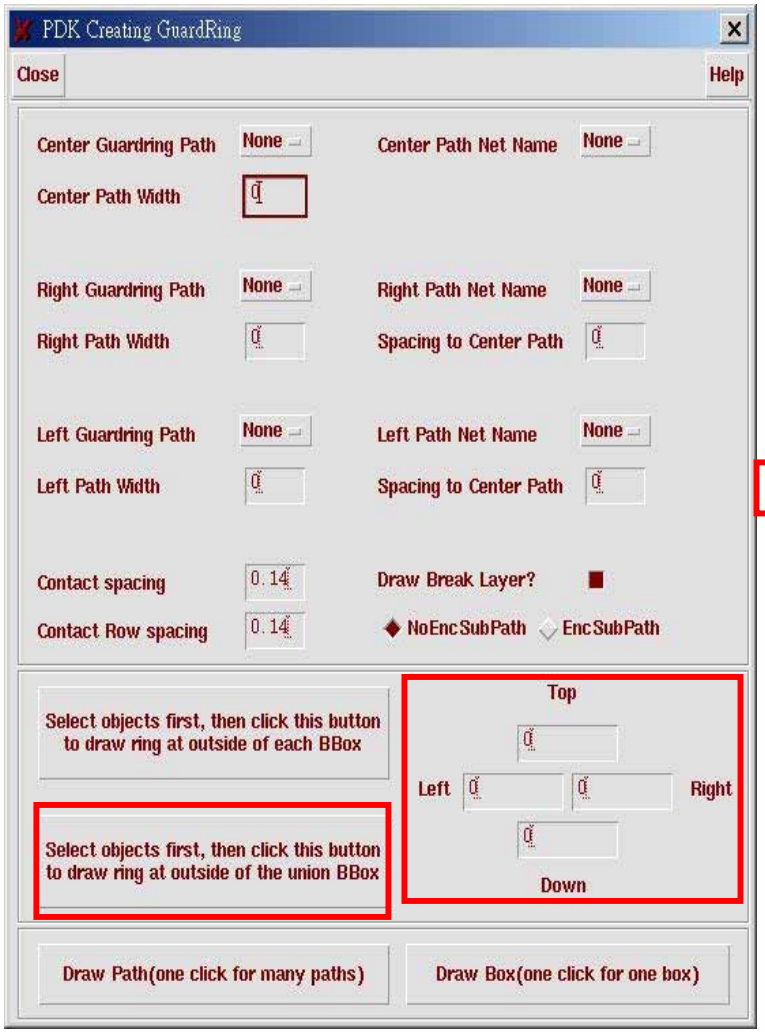
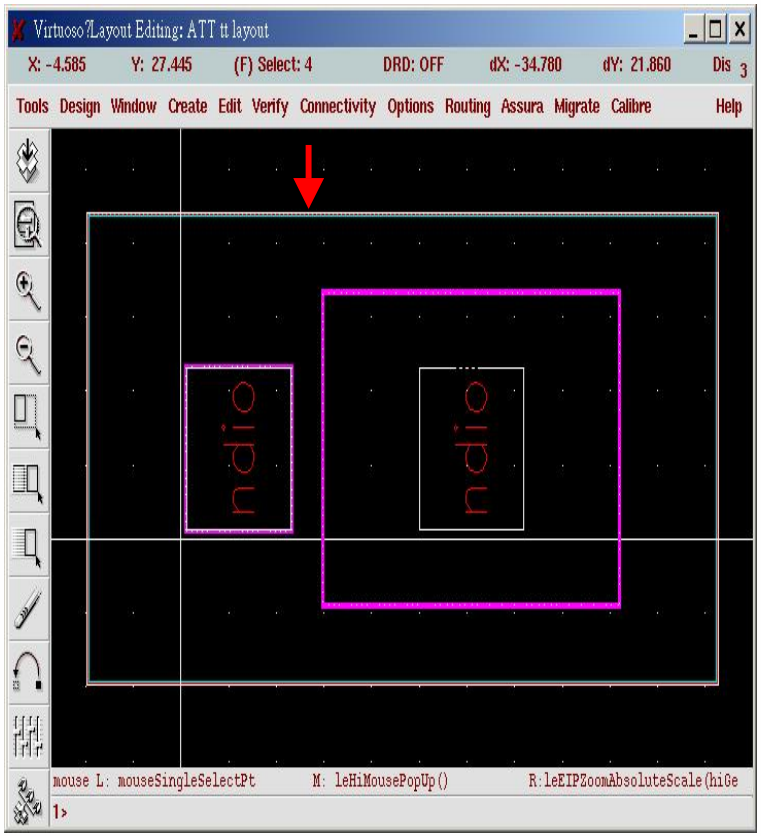
- **Create GuardRing utility :**





Create Guard-Ring outside of all instances by specified distance

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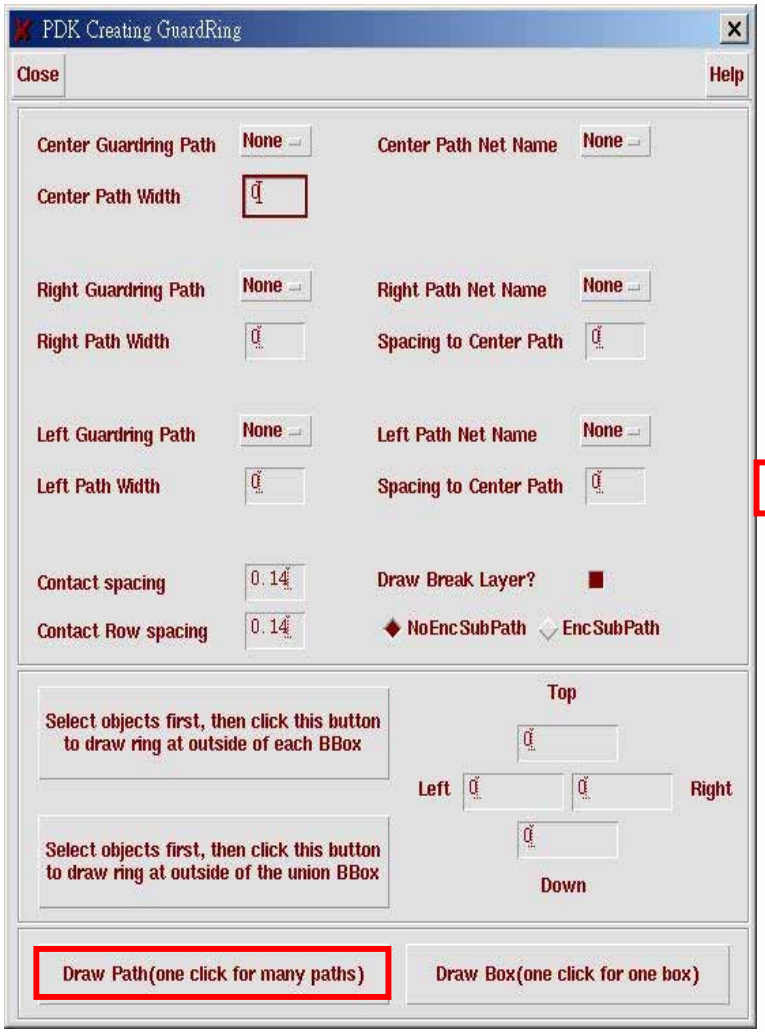
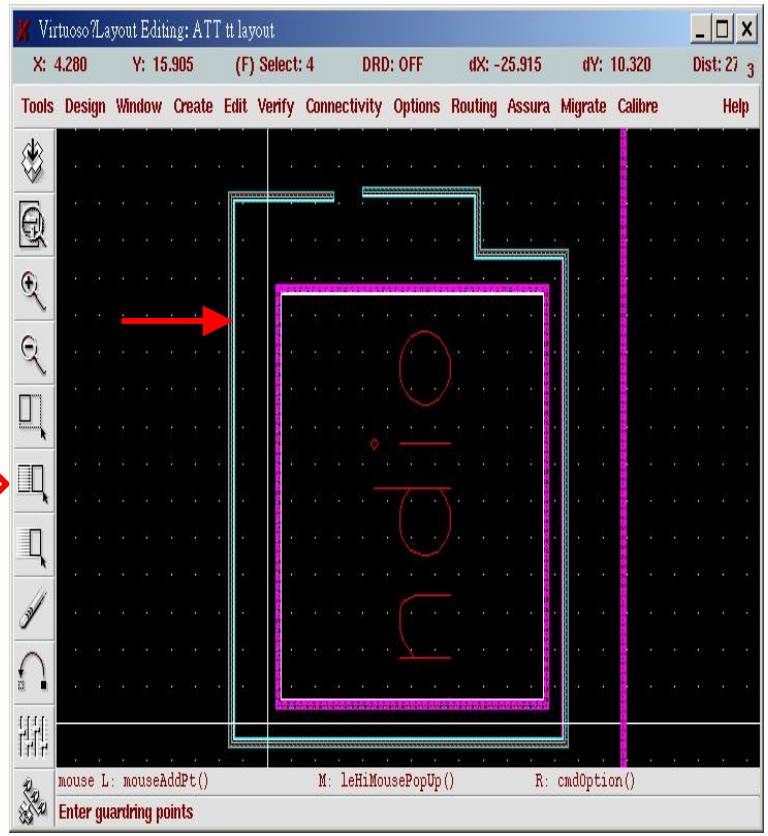
- **Create GuardRing utility :**





Create Guard-Ring path by hand

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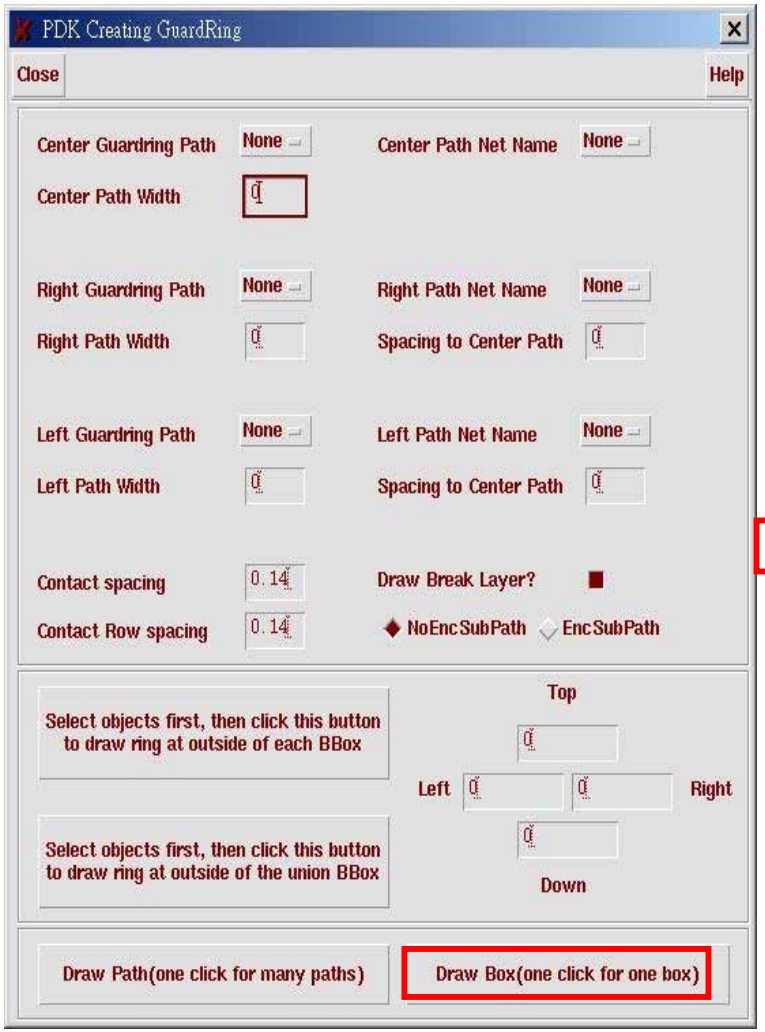
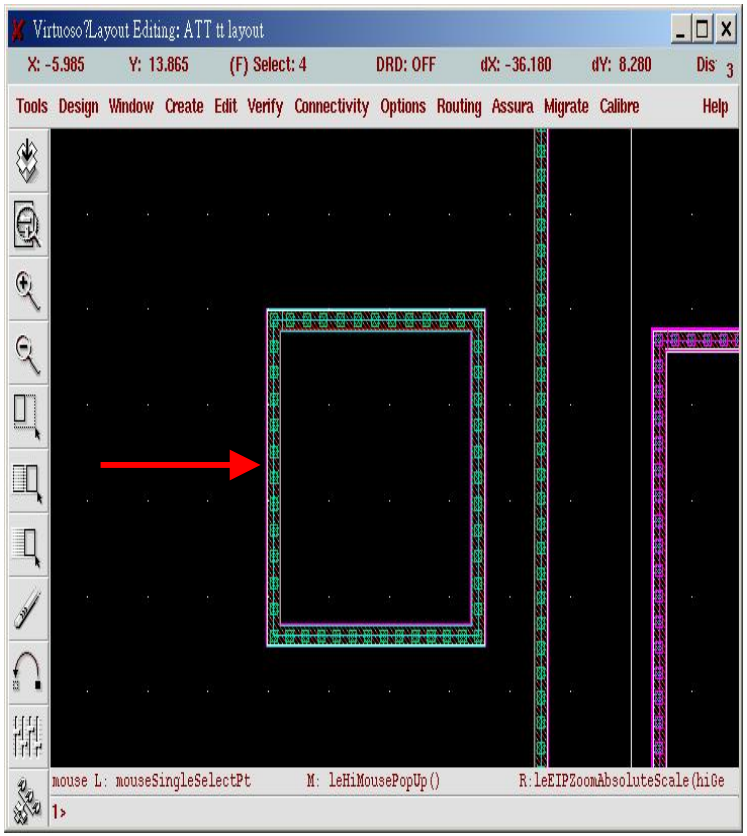
- **Create GuardRing utility :**





Create Guard-Ring box by click two points

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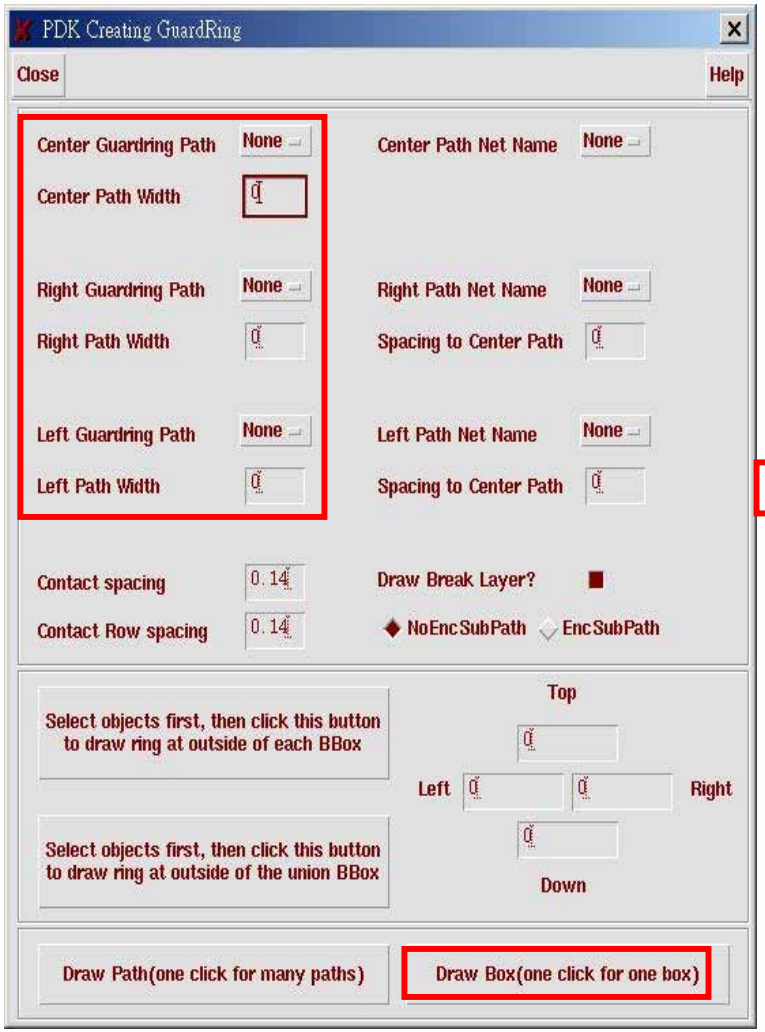
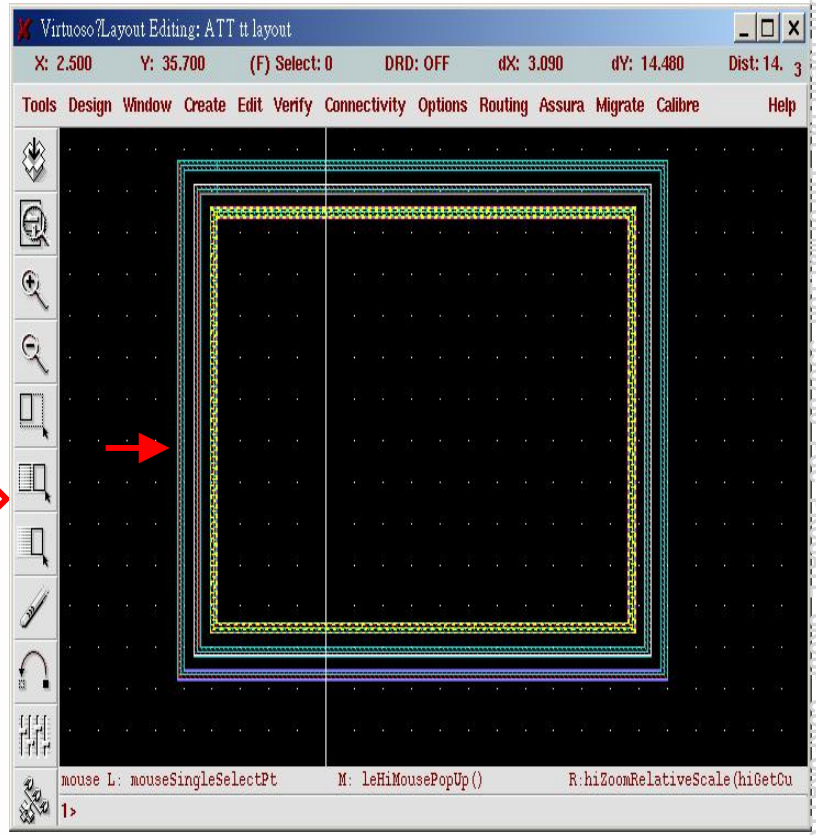
- **Create GuardRing utility :**





Create multi-Guard-Ring box by click two points

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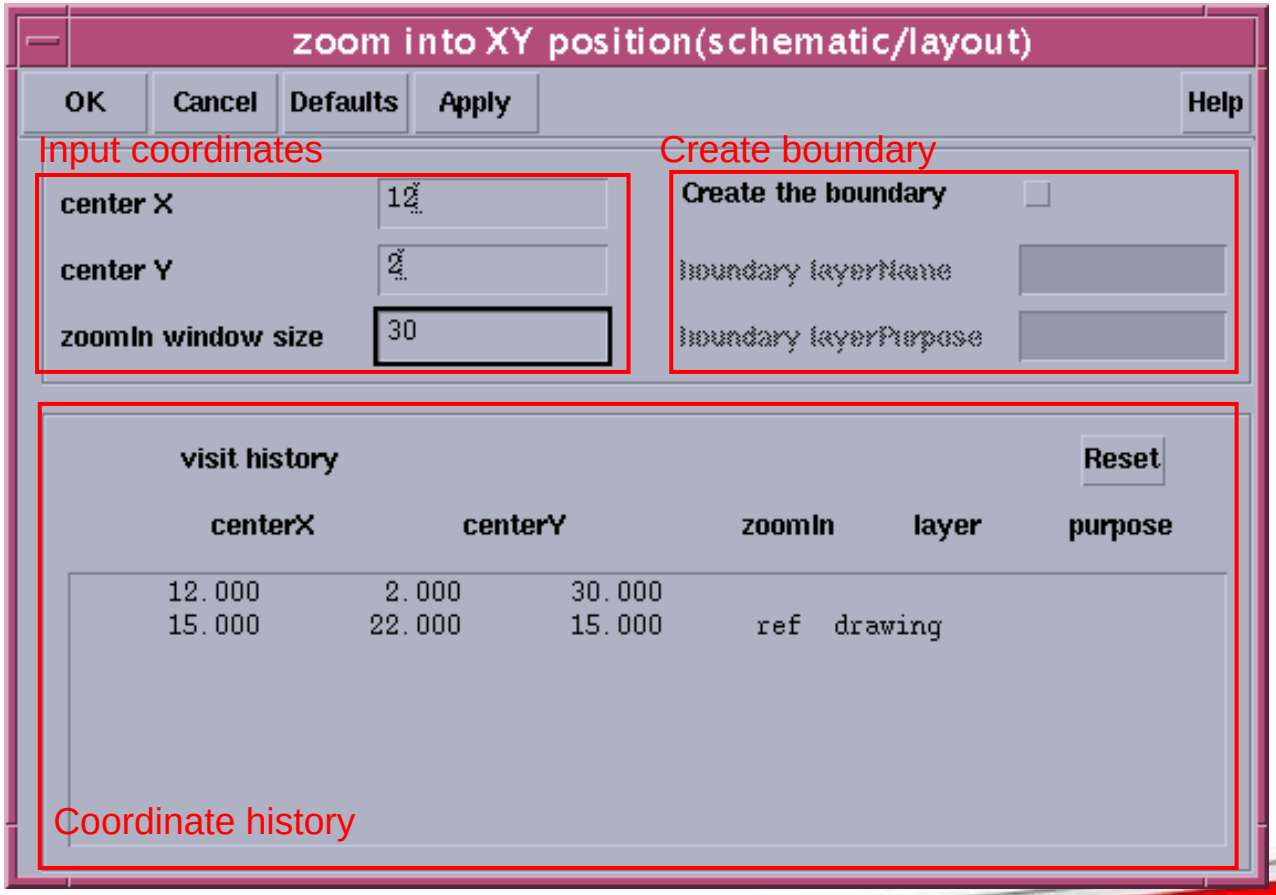
- **tsmcPdkZoomIn(sch/lay) with history :**
  - The function of tsmcPdkZoomIn(sch/lay) with history is an utility to help user easy to find the layout.



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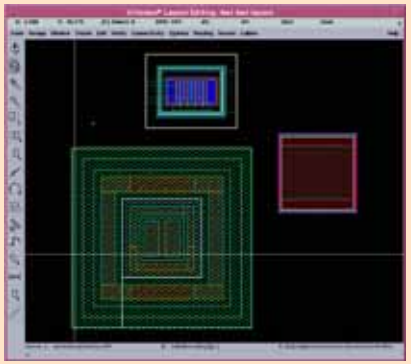
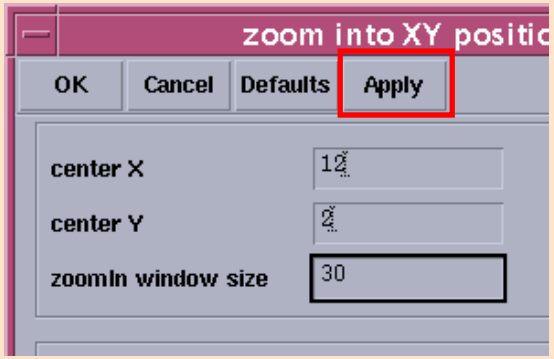
Way1

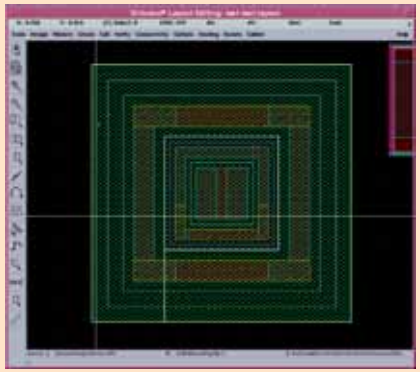
Select layout window



input coordinates and check Apply

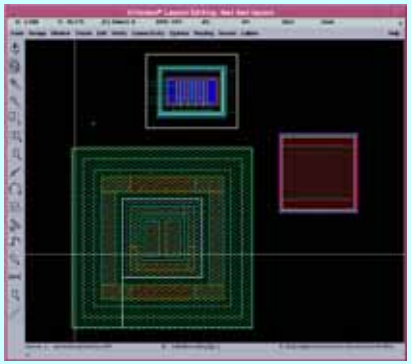


Window zoom in

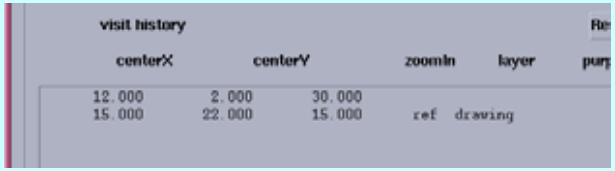


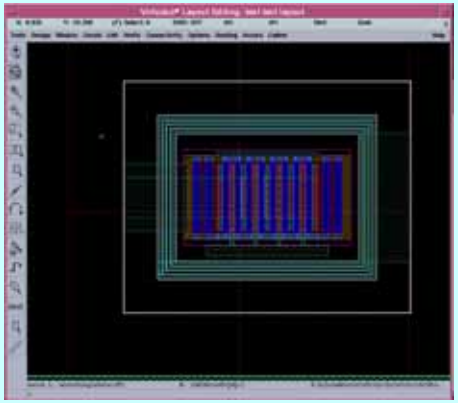
Way2

Select layout window



Double check coordinate history





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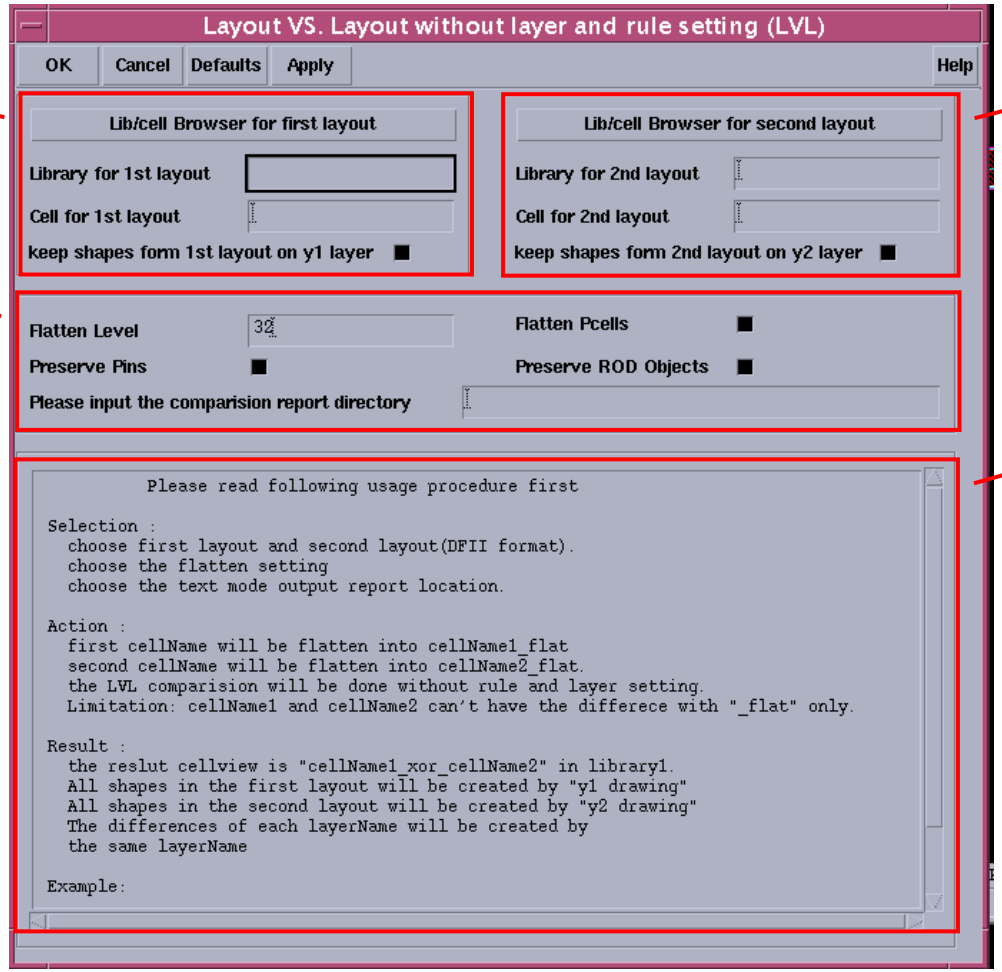
- **Layout vs. Layout :**
  - Layout vs. Layout (tsmcPdkLVL) is an utility to help user easy to compare the layouts.

First cell setup

Second cell setup

Flatten option

Function description



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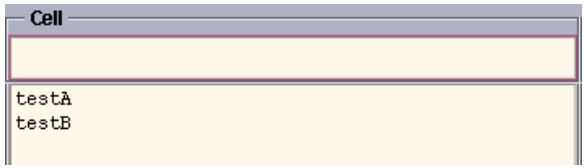
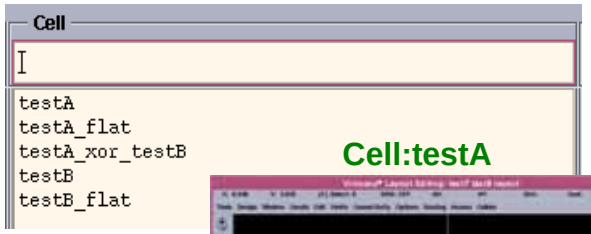
- **Layout vs. Layout :**

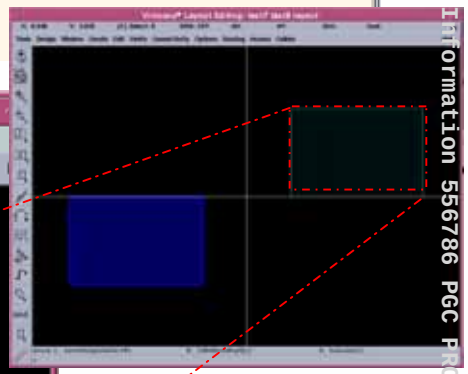
Before LVL



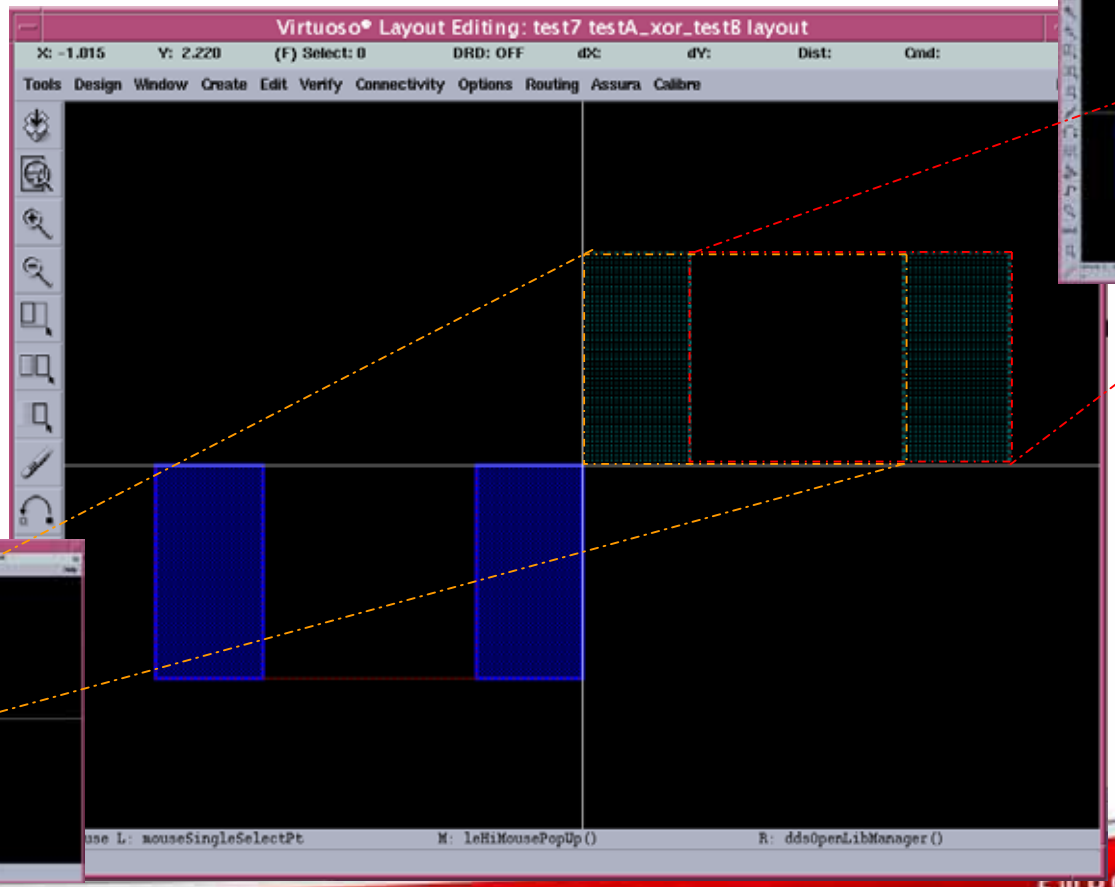
After LVL



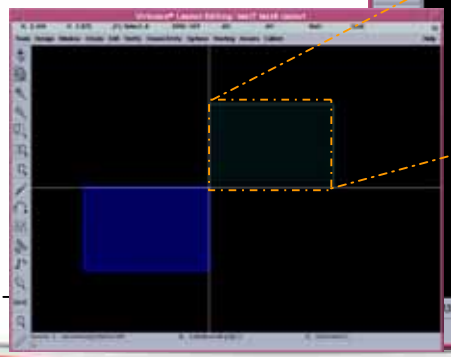
Cell:testA



Cell:testA_xor_testB



Cell:testB



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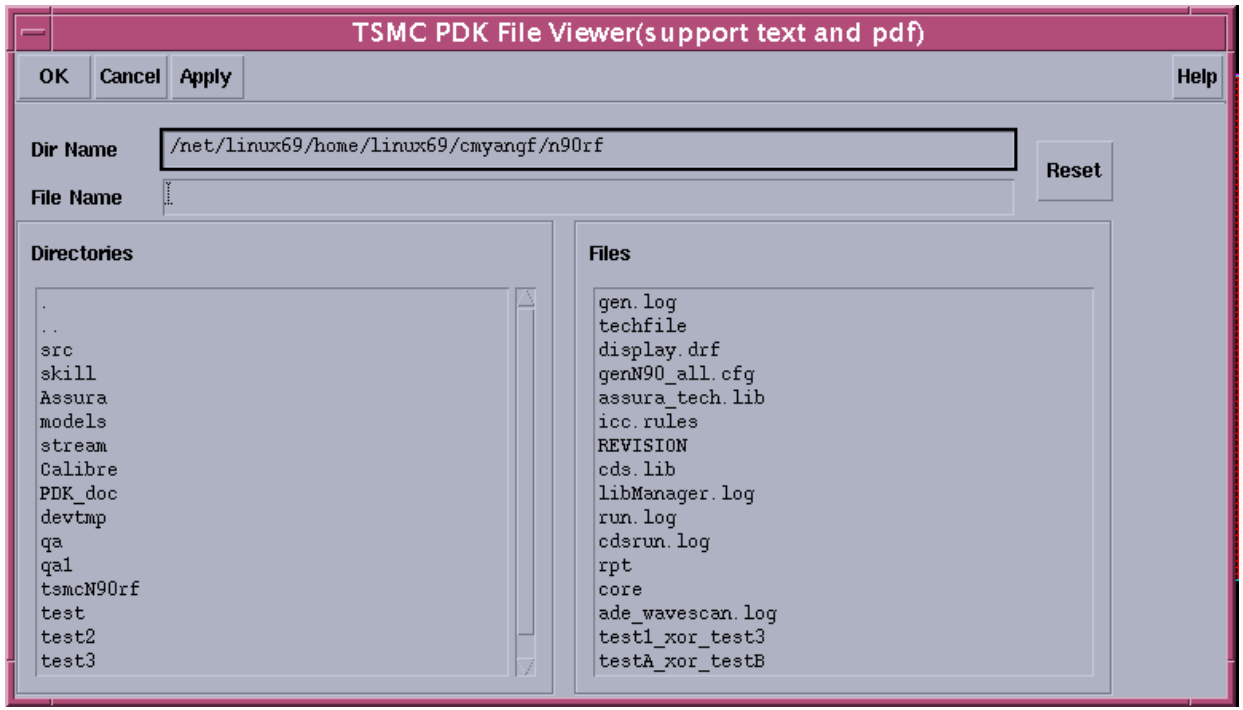
File Browser & Viewer :

- File Browser & Viewer – TSMC file browser tool, user can easily find the file they needs.



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- **tsmcPdkToolAbout :**

TSMC PDK setting information

Close | Help

```
---------------------
 (1) TSMC PDK information
---------------------
  Library : tsmc18
  Version : v1.1a

-------------------------
 (2) PDK installation information
-------------------------
*** Select mimcap :  1.0fF per unit area
*** Select package : TSMC 0.18um 1.8V/5V 1P6M Mixed Signal BSIM4.5 PDK
*** Install to directory : 1

--------------
 (3) PDK support
--------------
pdk_support@tsmc.com
```

	PDK	User
Operation System	Linux 2.4.21-47.ELsmp	Linux 2.4.21-47.ELsmp
DFII database	5.10.41.500.5.101	5.10.41.500.5.101
Assura version	317_USR2_HF7	317_USR2_HF7
Calibre version	v2007.3_36.25	v2007.3_36.25
Spectre version	6.2.1.264	6.2.1.264
Hspice version	Hspice < 2007	Hspice < 2007
Output to fileName	./tsmc18_pdk_info	Save As

View ReleaseNote | View REVISION

/net/linux197/home/linux197/pdk/alex/018mm_18v_5v_QA/PDK3/tsmc18/ /PDK_doc/ReleaseNote.txt

File | Help | 15

```
                TSMC 0.18um Mixed Signal SALICIDE 1.8V/5V PDK

                        Version : V1.1a
                        Doc. #  : T-018-CM-SP-010-K1
                        Date    : January 20, 2009

*****************************************************************************
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*****************************************************************************

0. Overview

 This PDK requires the following environmental variables:

       "CDS_Netlisting_Mode" to be set to "Analog"
       "CDSHOME" to be set to Cadence DFII installation path
```

/net/linux197/home/linux197/pdk/alex/018mm_18v_5v_QA/PD...

File | Help | 17

Revision	Date	Comments
1.1a	2009/1/20	New Create

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